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Impacts of the U.S. Department of Energy's Designation of South Korea as a Sensitive Country on the Science and Technology Sector and Response Measures

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Ⅰ Background and Current Status

- The South Korean government has engaged in consultations with the U.S. government through multiple channels to seek the removal of the designation as a "sensitive country,"* but the designation took effect on April 15, 2025.
  - However, the current Minister of Trade, Industry and Resources stated that, during a meeting with the U.S. Secretary of Energy, the two countries agreed to cooperate toward the removal of the sensitive country designation. Despite these diplomatic efforts, South Korea remains designated as a sensitive country.[1)]

    * Sensitive Country List (SCL): A list of countries to which particular consideration is given for policy reasons during the U.S. Department of Energy's (DOE) internal review and approval process for access by foreign nationals. (DOE O 142.3B, as of March 2022)

- The Sensitive Country List (SCL) is administered by the Office of Intelligence and Counterintelligence (OICI), an intelligence unit within the U.S. DOE.[2)] South Korea has been classified as an "Other Designated Country" on this list.
  - The primary restrictions include limitations on the sharing of advanced technological information with the U.S., including in fields such as nuclear energy and AI. Additionally, special approval is required for access to facilities operated by the U.S. DOE. Moreover, all U.S. federal employees are subject to restrictions when interacting with foreign nationals from sensitive countries.

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1) Dong-A Ilbo (March 22, 2025), "South Korea-U.S. agree to seek swift resolution of 'sensitive country' issue… Government says U.S. is positive about removal," https://www.chosun.com/international/us/2025/03/21/Z77BV2LJI5E53BSX6WVFCP3J6I/ (accessed: April 17, 2025)

2) OICI provides original scientific and technological intelligence and expertise to inform national security decision-makers and performs the duty of controlling threats to DOE organizations and to national energy security.

- The U.S. DOE has not disclosed the specific reasons for the sensitive country designation; however, it is presumed to be related to security concerns arising from the expansion of domestic nuclear-related research and the growth of related industries.
  - Domestically, President Yoon Suk Yeol has officially mentioned the possibility of possessing nuclear weapons,[3] and externally, the conclusion of a mutual defense treaty between Russia and North Korea has intensified discussions regarding South Korea's potential nuclear armament.[4]
  - Within the U.S., discussions have also taken place regarding potential developments such as South Korea's independent nuclear armament or nuclear sharing. However, toward the end of the Joe Biden administration, critical views and concerns regarding South Korea's potential acquisition of nuclear weapons began to spread.

    ※ Robert Peters (The Heritage Foundation): If South Korea were to possess independent nuclear weapons, it would pose a serious threat to the security situation in Northeast Asia.

    ※ Harry B. Harris Jr. (former U.S. Ambassador to South Korea): Independent nuclear armament by South Korea and Japan would generate significant tensions that could undermine peace in Northeast Asia.
  - The DOE's sensitive country designation should be understood not as a permanent measure but as a temporary step taken in response to specific circumstances.

    ※ In the past, the DOE designated South Korea as a sensitive country on two occasions–(1st) January 1986–September 1987 and (2nd) January 1993–June 1996–and the designation was lifted in 1994.[5]

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3) VOA (January 12, 2023) "South Korean President Yoon Suk Yeol mentions possibility of 'indigenous nuclear armament'… Presidential Office says remarks intended to signal firm response to North Korean nuclear threat, reaffirms commitment to the NPT," https://www.voakorea.com/a/6915173.html (accessed: April 19, 2025)

4) Future Korea (September 23, 2024), "[In-depth analysis] 'South Korea can no longer rely solely on the United States': A National Assembly–driven strategy for nuclear armament," https://www.futurekorea.co.kr/ews/articleView.html?idxno=147293\ (accessed: April 17, 2025)

5) Yonhap News (March 17, 2025), "South Korea previously placed on the U.S. sensitive country list and removed in 1994 (comprehensive)," https://www.yna.co.kr/view/AKR20250317115651504 (accessed:

- Records indicate that the first designation of a sensitive country dates to 1981, with a removal occurring in 1994.[6)]
- At the 1st South Korea–U.S. Joint Committee on Science and Technology held in December 1993, the Korean side requested that the U.S. lift the sensitive country designation. The removal was subsequently confirmed at the 15th South Korea–U.S. Joint Standing Committee on Nuclear Energy Cooperation held in Seoul in January 1994.[7)]

  ※ Korean side: Ministry of Science and ICT; U.S. side: Department of State
- At that time, the Korean side requested revisions to DOE internal regulations that required Korean researchers to undergo a series of screening procedures when visiting the DOE and its 19 affiliated laboratories, arguing that these procedures constituted a barrier to personnel exchanges between the two countries.

Following newspaper reports regarding the "sensitive country" designation, relevant government ministries initiated active consultations through various diplomatic channels to verify the facts, assess the background and potential impacts of the designation, and seek its removal.

- On March 10, 2025, an exclusive media report by Hankyoreh first revealed that DOE had been pursuing the designation of South Korea as a "sensitive country."
- Subsequently, the South Korean government verified the designation and worked to examine its background and circumstances while activating an interagency response mechanism aimed at removing the designation.
  - Relevant ministries, including the Ministry of Foreign Affairs, the Ministry of Science and ICT, and the Ministry of Trade, Industry and Resources, maintained

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April 17, 2025)

6) Hankook Ilbo (March 23, 2025), "Repercussions of the U.S. designation of South Korea as a 'sensitive country'… When did U.S. distrust and concern begin to emerge?" https://www.hankookilbo.com/News/Read/A2025032108160000568 (accessed: April 15, 2025)

7) Kyunghyang Shinmun (March 28, 2025), "Thirty years ago the U.S. designated 50 countries, including South Korea, as 'sensitive countries'… Concerns over nuclear policy cited as a possible reason,," https://www.khan.co.kr/article/202503281417001 (accessed: April 17, 2025)

ongoing communication and actively engaged in consultations with related U.S. institutions, including the White House, the DOE, and the Department of State.

- The U.S. side has reaffirmed that the recent DOE measure is unlikely to have a negative impact on South Korea-U.S. cooperation in science and technology. Both sides have discussed ways to strengthen collaboration in advanced technology sectors such as energy and AI.
  - On March 17, the spokesperson of the Ministry of Foreign Affairs stated that the background of the sensitive country designation was related to security issues at DOE laboratories and conveyed that the U.S. side had confirmed that the measure would not significantly affect technological cooperation, including joint research, between South Korea and the U.S.[8)]
  - In diplomatic circles, it has been suggested that the designation may have resulted from cases in which Korean researchers violated security regulations while visiting or conducting joint research at DOE laboratories.[9)]

    ※ An employee of a contractor at Idaho National Laboratory (INL) was detected attempting to board a flight to South Korea while carrying reactor design software classified as export-controlled information (between October 1, 2023 and March 31, 2024).[10)]
  - On March 19, at a meeting with representatives from government-funded science and technology research institutes, the First Vice Minister of Science and ICT stated that, according to confirmation received from the DOE regarding the impact of the "sensitive country" designation, there would be no problems for science and technology cooperation and that the U.S. continues to express strong willingness for future cooperation.[11)]
  - On March 20, the Minister of Trade, Industry and Resources held talks with the Secretary of Energy to discuss ways to strengthen bilateral cooperation across

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8) Ministry of Foreign Affairs Spokesperson's Regular Briefing (March 18, 2025)
9) Kyongbuk Ilbo (March 17, 2025), "Government mobilizes full response to the U.S. 'sensitive country' designation… Seeking Korea's removal from the list," https://url.kr/oupngl (accessed: April 18, 2025)
10) Hankook Ilbo (April 14, 2025), "U.S. 'Sensitive Country' list likely to take effect as scheduled on the 15th… What restrictions could South Korea face?" https://url.kr/wk6a1p (accessed: April 19, 2025)
11) Ministry of Science and ICT Press Reference Material (March 19, 2025)

various energy sectors*, as well as the South Korea-U.S. energy policy dialogue and the institutionalization of a public-private joint energy forum.[12]

* Including LNG, power grids, hydrogen, and nuclear energy such as SMRs

• **Table 1** • Timeline of Key Government Responses by Ministry Regarding the "Sensitive Country" Designation (2025)

Date	Ministry / Agency	Key Details
3.10.	Hankyoreh[13]	[Exclusive] U.S. DOE designates South Korea as a "sensitive country" and prepares follow-up measures
3.11.	Ministry of Foreign Affairs	Coordination with relevant ministries to verify the information
3.13.	Ministry of Foreign Affairs	Sensitive Country List not yet finalized; in close consultation with relevant U.S. agencies
3.17.	Ministry of Foreign Affairs	Official announcement: the background for the sensitive country designation is security issues at U.S. DOE laboratories
	Interagency Economic Issues Meeting[14]	Initiated development of cross-ministry response measures - Acting President Choi Sang-mok requested that "relevant agencies actively explain to the U.S. side to ensure there is no negative impact on South Korea-U.S. science, technology, and energy cooperation"
3.19.	Ministry of Science and ICT[15]	Held a meeting with government-funded research institutes to strengthen South Korea-U.S. science and technology cooperation - First Vice Minister Lee Chang Yune stated that "through dialogue with the DOE, it was confirmed that there are no issues with science and technology cooperation and that there is willingness for future collaboration" - Shared personnel and policy trends in science and technology under the new U.S. administration and discussed exploring new South Korea-U.S. cooperation items
3.20.	Ministry of Trade, Industry and Resources	South Korea-U.S. energy ministers' bilateral meeting - Agreed to proceed with negotiations to lift the "sensitive country" designation and cooperate for prompt resolution according to procedure

12) Ministry of Trade, Industry and Resources Press Reference Material (March 21, 2025)

13) Hankyoreh (March 10, 2024), "Exclusive: Amid growing calls for nuclear armament, the U.S. classifies South Korea as a 'sensitive country'… Concerns that cooperation in advanced technologies such as AI

Date	Ministry / Agency	Key Details
4.1.	Ministry of Science and ICT[16)	Minister Yoo Sang-im held a video call with Michael Kratsios, Director of the White House Office of Science and Technology Policy, shortly after his inauguration - Reaffirmed that Korea is a key U.S. partner and requested support to ensure that ongoing South Korea-U.S. science and technology cooperation between governments and research institutions continues without disruption
4.3.	Ministry of Foreign Affairs[17)	South Korea-U.S.-Japan Foreign Ministers' Meeting held and joint statement issued - Emphasized the need for joint efforts and leveraging each country's industrial capabilities to develop and deploy advanced civilian reactors to meet growing energy demand
4.14.	Ministry of Foreign Affairs[18)	Minister Cho Tae-yul met with U.S. Congressional Study Group on Korea (CSGK) delegation - Requested sustained attention from the U.S. Congress to strengthening South Korea-U.S. economic cooperation in shipbuilding, energy, advanced technology, and AI
4.15.	Ministry of Foreign Affairs	U.S. DOE's sensitive country designation takes effect (4.15.) - During ongoing director-level South Korea-U.S. working-level consultations aimed at lifting the designation, it was reconfirmed that current and future South Korea-U.S. R&D cooperation will not be affected

Source: Prepared by STEPI researchers with reference to Hankyoreh (March 10, 2025), Ministry of Foreign Affairs Spokesperson's Regular Briefing (March 11, March 13, March 17, April 15, 2025), Yonhap News (March 17, 2015), Ministry of Science and ICT Press Release (March 20, 2025), Ministry of Trade, Industry and Resources Press Reference Materials (March 21, 2025), Ministry of Science and ICT Press Statement (April 1, 2025), Ministry of Foreign Affairs Press Release (April 3 and April 14, 2025)

could be blocked," https://www.hani.co.kr/arti/politics/politics_general/1186120.html (accessed: April 18, 2025)

14) Yonhap News (March 17, 2025), "Choi: 'Industry Minister to hold talks with U.S. Energy Secretary this week'… Emphasizing response to the 'sensitive country' designation," https://url.kr/3x3k32 (accessed: April 19, 2025)

15) Ministry of Science and ICT Press Release (March 20, 2025)

16) Ministry of Science and ICT Press Statement (April 1, 2025)

17) Ministry of Foreign Affairs Press Release (April 3, 2025)

18) Ministry of Foreign Affairs Press Release (April 14, 2025)

- As confirmed through the background and current status of the sensitive country designation, the issue is recognized in the context of nuclear security and as being closely linked to science and technology cooperation (R&D).
  - Although it was reconfirmed that DOE's measures would not have a negative impact on South Korea-U.S. science and technology cooperation, it is necessary to examine the potential effects of this case on South Korea's science and technology community.
    - As South Korea-U.S. R&D cooperation is a key development strategy for advancing South Korea's Twelve Critical and Emerging Technologies, any restrictions on exchanges between Korean and U.S. scientists resulting from the sensitive country designation carry significant implications for South Korea's science and technology community.
  - Even if the sensitive country designation is expected to have minimal impact on South Korea-U.S. science and technology cooperation, efforts are needed to identify potential constraints arising from it and explore appropriate countermeasures.
    - In particular, in preparation for similar cases involving the U.S. or other countries in the future, such investigations and analyses may serve as valuable groundwork for establishing a proactive response system.
- Therefore, a balanced approach is required to avoid overinterpreting the sensitive country designation issue and misrepresenting its essence while preparing preemptive measures for anticipated challenges to the science and technology community.
  - Accordingly, this report analyzes factors that may hinder South Korea-U.S. science and technology community cooperation from the perspectives of ① research and strategic technology collaboration, ② export controls, and ③ expansion and spillover effects involving other U.S. agencies and proposes policy recommendations for addressing these challenges.

II Key Anticipated Issues (Impact on the Science and Technology Community)

1. Issues Related to Research and Strategic Technology Collaboration

- There may be an impact on science cooperation in emerging technology fields between the Ministry of Science and ICT and the National Nuclear Security Administration (under the DOE).[19)]
  - The Memorandum of Cooperation (MoC) signed by South Korea, the United States, and Japan in December 2023 includes an agreement to strengthen joint R&D collaboration in priority core and emerging technology areas between DOE laboratories and South Korean research institutions.
    - Notably, this agreement operationalizes the political understanding discussed at the South Korea-U.S.-Japan leaders' summit at Camp David in August 2023 and was signed prior to the South Korea-U.S.-Japan national security advisors' meeting.
  - The DOE's sensitive country designation represents a significant constraint on South Korea-U.S. science and technology development cooperation, and it may pose a burden for future joint development of core technologies.
  - Research cooperation in future technology areas such as "next-generation nuclear reactors" and "spent nuclear fuel reprocessing" requires careful examination of potential impacts, as most of these collaborations occur with DOE laboratories.
    - In particular, in U.S.-led technology areas such as fourth-generation SMRs (small modular reactors), it is necessary to assess practical issues affecting bilateral technology development cooperation, such as possible restrictions on U.S. site

19) DOE (December 8, 2023), "United States, Japan, and ROK sign trilateral framework encouraging scientific cooperation in CETs," https://www.energy.gov/nnsa/articles/united-states-japan-and-republic-korea-sign-trilateral-framework-encouraging?utm_source=chatgpt.com (accessed: April 13, 2025)

visits or the inability to import research equipment into South Korea.[20)]

※ There is additional ongoing research under the South Korea–U.S. Joint Fuel Cycle Study (JFCS) on pyroprocessing, a technology related to "spent nuclear fuel reprocessing."

- Furthermore, research collaborations on renewable energy and computational science between KIST and Lawrence Livermore National Laboratory,[21)] as well as joint research on next-generation secondary batteries involving KIST, Argonne National Laboratory, and Brookhaven National Laboratory,[22)] may also be affected.

Table 2 Status of South Korea–U.S. Research Institute and University Cooperation MOUs (MOUs valid as of the end of February 2025)

South Korean Research Institute	U.S. Research Institute
Korea Institute of Science and Technology (KIST)	Argonne National Laboratory, University of Minnesota, etc. (6 total)
Electronics and Telecommunications Research Institute (ETRI)	National Institute of Standards and Technology, Virginia Tech, etc. (6 total)
Korea Institute of Industrial Technology (KITECH)	Ames National Laboratory, Purdue University, etc. (4 total)
Korea Institute of Oriental Medicine (KIOM)	University of Maryland School of Medicine, etc. (2 total)
Korea Research Institute of Bioscience and Biotechnology (KRIBB)	University of Pennsylvania
Korea Institute of Science and Technology Information (KISTI)	Harvard University
Other 17 government-funded research institutes	47 international cooperation MOUs

Source: Maeil Business Newspaper (March 17, 2025) "Yellow card for Yoon Suk Yeol's 'nuclear armament theory'?… Biden drives a final nail with 'sensitive country' designation just before leaving office," https://www.mk.co.kr/news/politics/11265235 (accessed: April 20, 2025)

20) The Chosun Daily (March 17, 2025), "Placing South Korea on the 'Sensitive Country' list alongside North Korea, China, and Russia… is the U.S. moving to check next-generation nuclear power plants?" https://www.chosun.com/politics/diplomacy-defense/2025/03/17/2OLOQNDUFRF4JDZMAJO6O5E6JQ/ (accessed: April 15, 2025)

21) KIST(August 16, 2022), "KIST strengthens research cooperation with Lawrence Livermore National Laboratory in the U.S.," https://kist.re.kr/ko/news/press-release.do?mode=view&articleNo=8567 (accessed: May 12, 2025)

22) Ohmynews (October 22, 2024), "South Korea and U.S. vice ministers for science discuss measures to strengthen cooperation in basic research and strategic technologies," https://www.ohmynews.com/NWS_Web/View/at_pg.aspx?CNTN_CD=A0003072783 (accessed: May 12, 2025)

The designation may act as a potential obstacle to South Korea-U.S. joint research and technological cooperation in the CET field.

- Concerns have been raised regarding problems that could arise from regulations newly applied as a result of the DOE's sensitive country designation.
  - Under 10 CFR Part 810, when a country is designated as a sensitive country,[23] activities previously subject to general authorization (e.g., non-sensitive nuclear technologies) may become subject to specific authorization procedures, and additional review processes could be adopted.

    ※ This could also imply that activities already requiring specific authorization (e.g., enrichment, reprocessing, plutonium fuel, and heavy water production) may be subject to even stricter regulations-more complex procedures may be required in the long term than in the past.
  - According to DOE O 142.3B, "Unclassified Foreign National Access Program,"[24] access to the national laboratories, technologies, and information under the DOE's National Nuclear Security Administration (NNSA) requires enhanced identity verification (Indices Checks), and requests for access must be submitted 45 days in advance.[25]
  - In 2024, researchers from 12 government-funded research institutes under the Ministry of Science and ICT officially visited 9 national laboratories under the DOE.

    ※ Over the past 5 years, the annual number of visits declined sharply during the COVID-19 pandemic-from 12 in 2020 to 0 in 2021-but rebounded to 56 in 2022, 48 in 2023, and 100 in 2024, indicating a rapid recovery in South Korea-U.S. science and technology exchanges.[26]

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23) U.S. Department of Energy, 'e810 Reporting', https://www.energy.gov/nnsa/10-cfr-part-810 (accessed: April 15, 2025)
24) DOE (January 21, 2021), DOE O 142.3B: Unclassified Foreign National Access Program. https://www.energy.gov/nnsa/articles/ib-1423-28-rev-3-unclassified-foreign-visits-and-assignments (accessed: April 19, 2025)
25) National Nuclear Security Administration, NNSA
26) inews24 (April 18, 2025), "Meetings between South Korean and U.S. researchers have increased… what now after the sensitive country designation?" https://www.inews24.com/view/1835648 (accessed: April 20, 2025)

- Considering the aforementioned DOE regulations, this could act as an obstacle to CET cooperation, particularly with respect to bilateral personnel exchanges and collaborative research activities.
- This is because CET constitutes a key component of U.S. national security, and the DOE Sensitive Country List definition specifies countries that may affect U.S. national security.[27][28]

※ Procedural delays may arise in cooperation in areas central to the U.S.-China technological rivalry, such as AI and quantum technologies, and there are concerns that additional monitoring procedures could be imposed even after project approval.

• In 2024, the DOE established the Office of Critical & Emerging Technologies* to strengthen research in AI, quantum information technology, biotechnology, and semiconductors (microelectronics).

* The Office of Critical & Emerging Technologies is a newly established DOE department responsible for leading R&D and policy coordination in advanced technology fields. It includes a Chief AI Officer and aims to build world-leading AI infrastructure through the Frontiers in Artificial Intelligence for Science, Security, and Technology (FASST)[29] initiative.

- (Semiconductors) The Microelectronics Science Research Centers (3 centers planned, covering areas such as materials, system design, and manufacturing science) will comprise 16 projects organized in a networked structure among research institutes and conducted across 10 national laboratories.[30]

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27) Paul Weiss (March 7, 2024), "White House Updates List of Critical and Emerging Technologies That Are Significant to National Security at Overview," https://www.paulweiss.com/insights/client-memos/white-house-updates-list-of-critical-and-emerging-technologies-that-are-significant-to-national-security (accessed: April 15, 2025)
28) DOE (January 15, 2021), "Sensitive Country List," https://www.directives.doe.gov/terms_definitions/sensitive-country-list (accessed: April 15, 2025)
29) DOE (November 11, 2024), "Frontiers in Artificial Intelligence for Science, Security and Technology (FASST)," https://www.energy.gov/fast (accessed: April 24, 2025)
30) DOE (December 23, 2024), "Department of Energy Announces $179 Million for Microelectronics Science Research Centers," https://www.energy.gov/science/articles/department-energy-announces-179-million-microelectronics-science-research-centers (accessed: April 24, 2025)

- The Office of Science is currently conducting various projects with 17 DOE national laboratories.

> - (Quantum) DOE's National QIS (Quantum Information Science) Research Centers[31] include Argonne National Laboratory's "Q-NEXT: Next Generation Quantum Science and Engineering" and Brookhaven National Laboratory's "C2QA: Co-design Center for Quantum Advantage," as well as collaborating research institutes across five fields, including Fermilab, Lawrence Berkeley National Laboratory, and Oak Ridge National Laboratory.
> - (BIO) Within the Office of Science, the Biological and Environmental Research (BER) Program[32] supports energy security and sustainable infrastructure development through predictive understanding of complex biological, Earth, and environmental systems.

- According to the DOE FY 2025 Budget in Brief released in 2024, DOE's AI research and activities are described in the budget overview section "Invests in Critical and Emerging Technologies"[33] as follows.

> *"DOE's AI-related activities include fundamental research and development of AI and the application of AI tools for machine learning while assuring the safety, security, and robustness of AI systems. DOE will also apply AI technologies to the nuclear security mission, including early detection of foreign nuclear proliferation activities."*

  - Thus, beyond the research dimension of AI, the DOE also explicitly highlights security-related applications, such as early detection through the application of AI technologies within a nuclear security framework.

- Accordingly, it is necessary to recognize that the present issue could extend beyond the nuclear and nuclear energy sectors and potentially spread to research related to strategic technologies from a security perspective.

31) Office of Science (DOE), "National Quantum Information Science Research Centers," https://nqisrc.org
32) Office of Science (DOE), "About the BER Program," https://www.berscience.org/about-ber
33) DOE (2025), "Department of Energy FY 2025 Budget in Brief," https://www.energy.gov/sites/default/files/2024-03/doe-fy-2025-budget-in-brief-v2.pdf (accessed: April 23, 2025)

- In the long term, it is also necessary to consider the possibility that the issue may extend to CET-related research areas-such as AI, quantum technologies, and biotechnology-conducted across various DOE offices and national laboratories.
- Moreover, diverse forms of exchange exist among national laboratories, universities, and industry across a range of areas, including DOE's basic scientific research and technology transfer. Accordingly, discussions should begin on developing new research security and protection procedures within South Korea-U.S. relations (given South Korea's designation as a sensitive country), moving beyond the existing framework of bilateral joint research.

2. Issues Related to Export Controls

There are concerns that strengthened export control measures related to "AI and advanced semiconductor technologies," continuing from the Biden administration into the second Trump administration, could increase rigidity in South Korea–U.S. science and technology cooperation.

- Beginning in October 2022, the Biden administration announced four rounds of broad export control measures* aimed at restricting China's access to AI and advanced semiconductor technologies.

  * These measures included export restrictions on new equipment to 140 Chinese companies, expanded application of the Foreign Direct Product Rule (FDPR), and regulations imposing restrictions on emerging technology areas such as high-bandwidth memory (HBM).[34]

    ※ The U.S. government also sought to prevent semiconductor equipment produced in other countries from being supplied to China. Malaysia, Singapore, Israel, Taiwan, and South Korea were included among the countries subject to the application of the FDPR.

34) Reuters (December 3, 2024), "US targets China's chip industry with new restrictions," https://www.reuters.com/technology/us-targets-chinas-chip-industry-with-new-restrictions-2024-12-02/ (accessed: April 18, 2025)

- In South Korea's case, the "new U.S. export controls on advanced AI (January 2025)"* classify the country as a core ally (Tier 1), allowing the use of AI semiconductors and models without restrictions.
  * The proposal revises the Export Administration Regulations (EAR) to strengthen existing export control measures on advanced AI chips and prevent circumvention of export controls.[35)]
  - If South Korea were downgraded to Tier 2 amid the rapidly changing international environment, growing sensitivity over technology transfer, and supply chain restructuring strategies, prior approval from the U.S. government could become necessary for the distribution, procurement, and supply of advanced semiconductors, equipment, and software.[36)]
  - In particular, if the Foreign Direct Product Rule (FDPR) were applied, products manufactured using U.S.-origin technology could also become subject to indirect export controls.
  - Additionally, indirect disadvantages could arise in areas such as conducting joint research with U.S. research institutes and attracting overseas investment.
  - Sensitivity surrounding technology transfer has intensified especially in the AI and semiconductor sectors, and such developments-combined with export control policies-could negatively affect South Korea's technological cooperation environment.
- Although the DOE's current measure is unlikely to affect export controls directly, careful review will be required going forward. When the DOE updated the Unclassified Foreign National Access Program* in 2022, export control provisions were added to the section outlining its primary objectives.
  * The update included export control provisions within the regulatory framework related to the DOE's sensitive country designation. (Excerpt: "A risk-based review and approval

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35) Ministry of Trade, Industry and Resources (January 14, 2025), "U.S. announces new export control measures for advanced artificial intelligence (AI)" https://www.motie.go.kr/kor/article/ATCL3f49a5a8c/170028/view (accessed: April 10, 2025)

36) U.S. Department of Commerce Bureau of Industry and Security, "Export Administration Regulations (EAR)," https://media.bis.gov/regulations/ear (accessed: April 18, 2025)

process for foreign national access consistent with United States (U.S.) law; national and economic security; and DOE program-specific policies, requirements, and objectives.” DOE O 142.3B, March 2, 2022)

- A risk-based review is conducted for foreign nationals seeking access to DOE facilities or participation in joint research.
- Researchers from sensitive countries are subject to additional approval procedures.
- Approval is granted only if the request complies with U.S. export control laws, national security policies, and DOE internal guidelines.

U.S. export control measures are closely linked to national security, and their scope and intensity may expand in response to changes in the geopolitical environment.

- In 2023, the Disruptive Technology Strike Force was launched to address illicit exports and leakage of advanced technologies, as well as threats to national security, through interagency cooperation.
  - The initiative is led by the U.S. Department of Justice (DOJ) and the Department of Commerce’s Bureau of Industry and Security (BIS). Working with agencies such as the Federal Bureau of Investigation (FBI) and Homeland Security Investigations (HSI) in major U.S. cities, its activities include preventing overseas leakage and misuse of advanced technologies, protecting supply chains, and investigating violations of export control regulations.[37)]
  - From a long-term security perspective, this development suggests the possibility that stricter export control enforcement could affect South Korea-U.S. joint research, personnel exchanges, the growing sensitivity surrounding advanced technologies, and supply chain restructuring in CET fields.

37) U.S. Department of Justice (February 16, 2024), “FACT SHEET: Disruptive Technology Strike Force Efforts in First Year to Prevent Sensitive Technology from Being Acquired by Authoritarian Regimes and Hostile Nation-States,” https://www.justice.gov/archives/opa/pr/fact-sheet-disruptive-technology-strike-force-efforts-first-year-prevent-sensitive (accessed: April 21, 2025)

3. Issues Related to Expansion and Spillover Effects Involving Other U.S. Agencies

The U.S. maintains a system in which individual federal departments independently designate and manage sensitive countries. As such departmental measures are influenced by national policies and strategies led by the White House, careful monitoring of these developments is required.

- Major federal departments designate sensitive countries in accordance with their respective laws and policies and impose stringent restrictions on technology transfer, cooperation, investment, and exports involving these countries.

※ In addition to the Sensitive Country List applied at the federal government-wide level, the U.S. federal government operates separate sensitive country designation systems depending on the role and mission of each federal department.

- Although they differ in purpose and scope of application, the sensitive country designation systems have direct effects on technology protection, research collaboration, export controls, and national security reviews.

Table 3 Sensitive Country-Related Designation Systems Operated by U.S. Federal Departments

Category	System Name or Criteria	Key Details
Department of Energy (DOE)	• Sensitive and Other Designated Countries List (SCL)	• Identifies countries requiring special consideration for policy reasons such as national security and nuclear proliferation concerns • Used in restrictions related to nuclear technology transfers, cooperation with national laboratories, and participation of foreign researchers in projects
Department of State, Department of Defense (DOS, DOD)	• ITAR §126.1 Proscribed Countries	• Restricts exports of defense technologies and materials to countries subject to sanctions under the International Traffic in Arms Regulations (ITAR)

38) U.S. Department of Commerce Bureau of Industry and Security, 'Supplement No. 1 to Part 740-Country

Category	System Name or Criteria	Key Details
Department of Commerce (DOC)	• Country Groups (A, B, D, E) under the Export Administration Regulations (EAR)[38] • Commerce Control List (CCL) • List of Concerned Parties:[39] Denied Person List, Entity List,[40] Unverified List (UVL), The Military End-User (MEU) List,[41] Consolidated Screening List	• Manages lists of countries, entities of concern, and controlled items that serve as the basis for restrictions on dual-use technology exports, reexports, and licensing requirements
Department of the Treasury's Office of Foreign Assets Control (OFAC)	• Specially Designated Nationals List (SDN List)[42]	• Imposes sanctions such as restrictions on financial transactions, exports and imports, investment, and asset freezes on lists of designated individuals, corporations, and organizations* * e.g., terrorists, drug traffickers
Office of Science and Technology Policy (OSTP)	• Malign foreign talent recruitment program prohibition (MFTRPs) (42 U.S.C. § 19232)	• Imposes restrictions on employment and project participation at U.S. federal research institutions for individuals or organizations involved in malign foreign talent recruitment programs* to protect research security and prevent technology leakage * e.g., DOE, NSF, NIH, DOD

Source: Prepared by STEPI researchers

Groups', https://www.bis.gov/regulations/ear/part-740/supplement-1-740/country-groups (accessed: April 17, 2025)

39) U.S. Department of Commerce Bureau of Industry and Security, 'Military End User List', https://www.bis.doc.gov/index.php/policy-guidance/lists-of-parties-of-concern/1770 (accessed: April 17, 2025)

40) U.S. Department of Commerce (February 5, 2025), 'Supplement No. 4 to Part 744; Entity List', https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-744/appendix-Supplement%20No.%204%20to%20Part%20744 (accessed: April 17, 2025)

41) U.S. Department of Commerce, Supplement No. 7 to Part; 'Military End-User' (MEU) List https://www.ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-744/appendix-Supplement%20No.%207%20to%20Part%20744 (accessed: April 17, 2025)

42) U.S. Department of the Treasury, Specially Designated Nationals List (SDN List), https://sanctionslist.ofac.treas.gov/Home/SdnList (accessed: April 17, 2025)

Additionally, to implement the national security policy (NSPM-33) announced at the end of the first Trump administration, the Office of Science and Technology Policy (OSTP) established guidelines that have since been incorporated into science and technology-related policies across federal agencies and are showing a trend toward broader expansion.

- On January 14, 2021, President Trump issued the "U.S. Government-Supported Research and Development National Security Policy (National Security Presidential Memorandum-33, NSPM-33)"[43] to prevent the leakage of U.S. science and technology and strengthen research security.
- On January 4, 2022, the Office of Science and Technology Policy (OSTP) released the NSPM-33 Implementation Guidance to operationalize the policy[44] and requested that all federal agencies establish and operate their research security policies while monitoring their implementation.
  - The detailed implementation guidelines specify the following measures: 1) standardization of information disclosure forms, 2) recommendation to adopt digital persistent identifiers, 3) establishment of research security programs, 4) sanctions for violations and information-sharing mechanisms (including prohibition of participation in MFTRP), and <u>5) strengthening of information-sharing and reporting protocols among federal agencies</u>.
  - In particular, institutions receiving more than $50 million in annual federal funding must establish research security programs, including cybersecurity, international travel security, insider threat mitigation, and export controls training as mandatory components.

43) The White House (January 14, 2021), "Presidential Memorandum on United States Government-Supported Research and Development National Security Policy (NSPM-33)". https://trumpwhitehouse.archives.gov/presidential-actions/presidential-memorandum-united-states-government-supported-research-development-national-security-policy (accessed: April 17, 2025)

44) National Science and Technology Council (NSTC), (2022.1). "Guidance for Implementing National Security Presidential Memorandum 33 (NSPM-33) on National Security Strategy for United States Government-Supported Research and Development," Office of Science and Technology Policy (OSTP)

- On January 19, 2024, the DOE established the "DOE Scientific Integrity Policy (DOE P 411.2B)" based on OSTP guidelines and NSPM-33.[45)]
  - The DOE, together with the NSF and NIH, participates in the Subcommittee on Research Security under the OSTP and has played a key role in designing OSTP implementation guidelines related to the national security strategy.
  - The policy established by the DOE includes research security obligations, the prohibition of MFTRPs, and rules governing external communications and publications. These provisions apply uniformly to all research institutions under the DOE.
- In addition to the DOE, other agencies-including the Department of Defense (DOD), the Department of Health and Human Services (including the NIH), the National Aeronautics and Space Administration (NASA), and the National Science Foundation (NSF)-are required to establish research security programs. Accordingly, each federal agency has developed and is implementing department-specific policies to comply with NSPM-33.

Considering the spillover effects among U.S. allied countries, it is necessary to examine the possibility that the designation of "other countries" may also lead to similar or linked regulations within U.S. allied countries and multilateral frameworks.

- Major countries outside the U.S. also operate a "Sensitive Country Lists" or comparable country classification systems as part of policies related to science and technology, export controls, and security.
  ※ Major countries maintain sanction lists primarily focused on countries of concern with respect to potential military end-use and weapons of mass destruction (WMD) and are expanding the scope of regulation beyond dual-use technologies and materials to include academic and research activities.
- Although countries and items subject to export controls are largely similar due to international nonproliferation agreements*, there is a growing trend toward expanding export controls and academic restrictions on non-strategic items at

45) U.S. Department of Energy (January 19, 2024), "Scientific Integrity Policy (DOE P 411.2B)," https://www.directives.doe.gov/directives-documents/400-series/0411.2-BPolicy-b (accessed: April 19, 2025)

the national strategic level as national-priority and protectionist policy orientations strengthen.

* Wassenaar Arrangement, NSG, MTCR, AG, etc.

- According to Article 22(5) of the Horizon Europe Regulation, for actions related to Union strategic assets, interests, autonomy or security, the work program may provide that the participation can be limited to legal entities established only in Member States or those established in specified associated or other third countries in addition to Member States.[46]
- Article 22(5) of the Horizon Europe Regulation has mainly been applied to work program topics related to quantum research, space, and critical raw materials.
  * The EU has restricted the participation of the UK, Switzerland, and Israel in major quantum and space research projects to protect strategic assets and technological sovereignty.[47]
  * The EU, citing concerns over technology leakage, military end-use, and unfair competition involving China, has restricted participation in strategic technology fields (high-performance computing, satellites, AI, and biotechnology) and market-proximate research within Horizon Europe while allowing only limited participation in environment-focused basic research.[48]
- Europe is establishing legal and institutional mechanisms that can restrict even academic cooperation from a national security perspective. Such restrictions on international cooperation are based on risk assessments* rather than discrimination against specific countries, with particular sensitivity in the area of critical technologies.[49]

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46) EC (June 29, 2023), "First biennial report on the implementation of the Global Approach to research and innovation," https://research-and-innovation.ec.europa.eu/news/all-research-and-innovation-news/first-biennial-report-implementation-global-approach-research-and-innovation-2023-06-29_en (accessed: April 19, 2025)
47) Kelly, E. (March 9, 2021), "Israel, Switzerland and UK face exclusion from major EU quantum and space research projects, Science|Business," https://sciencebusiness.net/news/quantum-computing/israel-switzerland-and-uk-face-exclusion-major-eu-quantum-and-space-research (accessed: April 12, 2025)
48) Open Access Government (2023.8.2.), "EU-China research thrives amid environmental focus," https://url.kr/3wxx1z (accessed: April 24, 2025)
49) EC (May 23, 2024.), "Council Recommendation - on enhancing research security (C/2024/33510)," Official Journal of the European Union.

* Among the 10 identified critical technology areas, risk assessments have initially been conducted in four areas: advanced semiconductors, AI, quantum technologies, and biotechnology.

Table 4 Sensitive Country Management Systems by Major Countries

<table>
<tr><th>Country</th><th>Major Institution</th><th>System Name</th><th>Purpose and Characteristics</th></tr>
<tr><td rowspan="2">EU</td><td rowspan="2">EC + Member States</td><td>• (Export controls) EU Dual-Use Control List</td><td>• Export and transfer controls related to dual-use technologies and materials
• Introduction of autonomous export controls (Catch-All) provisions, strengthened controls on technologies related to human rights violations (surveillance, cybersecurity, AI, biotechnology, etc.)</td></tr>
<tr><td>• (National sanctions) CFSP Sanctions*
* (Condition) Unanimous agreement of all EU Member States</td><td>• Political sanctions* targeting countries, organizations, and individuals for the protection of international peace, human rights, and democracy
* Asset freezes, trade restrictions, travel bans, etc.</td></tr>
<tr><td rowspan="2">UK</td><td>Department for Business and Trade (DBT)</td><td>• UK Strategic Export Control Lists (SECL)</td><td>• Technology-and product-based export controls on military equipment and dual-use technologies</td></tr>
<tr><td>Foreign, Commonwealth & Development Office (FCDO)</td><td>• UK Sanctions List (UKSL)</td><td>• Sanctions list targeting countries, individuals, and organizations for political, diplomatic, and security purposes</td></tr>
<tr><td rowspan="2">Japan</td><td rowspan="2">Ministry of Economy, Trade and Industry (METI)</td><td>• Foreign End User List (FEUL)</td><td>• Recipient-based controls on companies/institutions suspected of involvement in the development of WMD
• Introduction of voluntary restrictions on research cooperation with universities and research institutions</td></tr>
<tr><td>• Export Trade Control Order (ETCO)</td><td>• Technology- and product-based controls
• Includes autonomous export controls (Catch-All) provisions</td></tr>
</table>

Country	Major Institution	System Name	Purpose and Characteristics
Canada	Global Affairs Canada (GAC)	• Area Control List (ACL)	• Country-based controls on all exports
		• Export Control List (ECL)	• Item-based export controls on sensitive items* * AI, weapons, semiconductors, nuclear equipment, etc.
Australia	Department of Defense (DOD)	• Defense and Strategic Goods List (DSGL)	• Controls on military and dual-use technologies and materials (exports, technology transfer, research sharing, etc.)
China	Ministry of Commerce (MOFCOM), Ministry of Science and Technology (MOST)	• Unreliable Entity List (UEL)	• Sanctions targeting foreign companies and individuals for protecting China's national security, public interest, and domestic industries * Prohibition of import/export and investment, visa and entry restrictions, administrative measures, etc. • Primarily retaliatory or counter-sanction measures targeting companies that cooperate with sanctions or control measures imposed by foreign governments • A system similar to the U.S. Entity List
		• Catalogue of Technologies Prohibited or Restricted from Export (technology export controls list)	• Specifies a list of advanced technologies developed by China whose overseas transfer is restricted * AI algorithms, genomic analysis and editing, etc. • Prior government approval is required when foreign companies acquire Chinese technologies or conduct joint research
		• 管制物项清单(List of Controlled Items) (Strategic Items Control List)	• Strict screening of exports to sensitive countries (North Korea, U.S., Taiwan, etc.) • China independently designates controlled items* (technologies similar to those in international agreements) * Weapons, biotechnology, aerospace, AI, encryption technologies, etc.

Source: Prepared by STEPI researchers

- Although there are differences in the purposes and constraints of Sensitive Country List operations across countries, such systems may be compatible with or influenced by U.S. control regimes among U.S. allies.

※ For U.S. allies, sanctioned countries are jointly designated or restricted through dual-use export control frameworks (e.g., the Wassenaar Arrangement).

- After the U.S. Department of Commerce announced new export control measures (October 2022) to restrict China's access to advanced semiconductor technologies, a meeting was held on January 27, 2023 with Japan and the Netherlands, resulting in an informal trilateral agreement to restrict exports of advanced semiconductor manufacturing equipment (e.g., ASML, Nikon Corporation, and Tokyo Electron) to China.[50]

- Subsequently, for national security and technology leakage prevention purposes, Japan's Ministry of Economy, Trade and Industry (METI) announced export controls on 23 types of semiconductor manufacturing equipment in March 2023, while the government of the Netherlands announced in June 2023 plans to strengthen export controls on advanced DUV immersion lithography equipment produced by ASML, its domestic semiconductor equipment manufacturer.

- Moreover, as the U.S. exerts diplomatic pressure on allied countries with high dependence on the U.S. to adopt coordinated sanctions or restrictive measures, intelligence-sharing allies or countries engaged in joint weapons development* often align their policies with those of the U.S.

* Five Eyes, NATO member states, etc.

- China, in response to sanctions imposed by the U.S., is strengthening the Anti-Foreign Sanctions Law* and rapidly expanding legal measures to protect its domestic technologies and respond to foreign pressure.

※ To respond to foreign sanctions, China announced and implemented provisions of the "Anti-Foreign Sanctions Law*"[51] (March 23, 2025), which specify in detail the scope of

50) J. Zhang. et al. (February 28, 2023), "Japan and the Netherlands Agree to New Restrictions on Exports of Chip-Making Equipment to China," Mayer Brown. https://www.mayerbrown.com/en/insights/publications/2023/02/japan-and-he-netherlands-agree-to-new-restrictions-on-exports-of-chipmaking-equipment-to-china (accessed: April 24, 2025)

application, retaliatory measures, and procedures across 22 articles.

* The law specifies concrete countermeasures against institutions or individuals that impose sanctions on China, including seizure, detention, or freezing of "other types of assets," prohibition or restriction of "related transactions, cooperation, and other activities (education, science and technology, environmental protection, economy and trade, etc.)," and "other necessary measures (prohibition or restriction of import and export activities and investment, restrictions on residence, etc.)."

Table 5 Comparison of Sensitive Country Sanctions by Major Countries

Item	U.S.	Europe	Japan	China
Clarity of Lists	Multiple lists exist by sector	Centered on integrated export controls regulations	Low risk of overlap	Strong political and economic response factors
Academic Restrictions	Very active (DOE, DOD, etc.)	Restrictions beginning for some countries (UK, Israel, China, etc.)	Increasing moves toward restrictions	Focus on preventing technology leakage
Strength of Implementation	Strong legal enforcement	Variation among member states	Primarily administrative guidance (guidance prepared for SMEs)	Centered on political sanctions
Alignment with U.S. Standards	Reference country	Generally similar	Similar	Contrasting (based on its standards)

Source: Prepared by STEPI researchers

- If the designation of South Korea as an "other country" by the U.S. DOE is perceived negatively by U.S. allied countries, it could lead to consequences such as international reputational damage, restrictions on international research

51) State Council (March 24, 2025), "实施《中华人民共和国反外国制裁法》的规定," 国令第-803号. https://www.gov.cn/zhengce/content/202503/content_7015400.htm (accessed: April 24, 2025)

cooperation, reduced investment from international firms, and restrictions on admission or hiring by university research institutes; therefore, preparedness is necessary.

- Additionally, South Korea must prepare for this measure by the U.S. DOE and potential implications arising from China's Anti-Foreign Sanctions Law, which was enacted in response to sanctions imposed by the U.S. against China.

Ⅲ Response Strategies (Policy Recommendations)

- (Short-term response) Although the sensitive country designation by the U.S. DOE is limited to a departmental-level measure within the U.S. government, various issues have become intertwined, leading to concerns about over-securitization and complex issue-linkage situations. Therefore, it is necessary to separate these factors and identify measures that can facilitate the lifting of the designation in the shortest possible timeframe.
- (Mid- to long-term response) Simultaneously, this situation highlights structural limitations in the current response system, including ▲ an issue-specific response structure centered on relevant ministries, ▲ reactive responses implemented after issues arise, and ▲ limited access to practical and specialized information. Accordingly, mid- to long-term response strategies should be established in terms of ① an integrated response, ② a preventive system, and ③ the securing of intelligence.[52]

1. Short-term Response Strategy

Recommendation 1. (Rapid lifting) Pursue the earliest possible lifting of the designation by avoiding over-securitization and separating this issue from others.

- (Over-securitization) This issue tends to have been somewhat expanded in review from the perspective of security threats (risk) due to the specificity of the nuclear energy-related field. Thus, it is necessary to avoid over-securitization and focus on solutions from the perspective of R&D cooperation, which is the core of the issue.

52) In general, "intelligence" is used in contrast to "information" and can be understood in two ways: (1) information that is not publicly available and (2) higher-level knowledge and views. In this article, it is used in the sense of "security-related knowledge."

- In the process of reviewing the issue in connection with domestic and international environments, the perspective of a security threat has been incorporated, expanding the perceived scope of the issue in a form that expands the estimated impact beyond actual causes.
- In other words, although there is specificity in the nuclear energy-related field, it should fundamentally be examined from the perspective of R&D cooperation.

(Issue separation) Furthermore, the need for issue linkage has been raised due to concerns that this measure could have spillover effects into other issues between South Korea and the U.S.; however, in this case, an "issue separation strategy" is more appropriate to achieve a rapid resolution.

- Although there are opinions that it is necessary to review linkages with other pending issues between South Korea and the U.S., such as its impact on South Korea-U.S. tariff negotiations, this may increase the complexity of the issue and delay the timing of lifting the designation.
- Moreover, an "issue linkage strategy" is fundamentally compromise-based and, therefore, is not strategic-it would require offering concessions ("quid pro quo") on the linked issues to achieve the lifting of the designation.
  - ☞ **In conclusion, for short-term response, it is necessary to recognize that the core issue is that a temporary disruption has occurred in the process of R&D cooperation and pursue a "one-point strategy" by separating it from other issues between the two countries to achieve a rapid resolution.**

2. Mid- to Long-Term Response Strategy

Recommendation 2. (Integrated response) Strengthening response capacity through establishment of whole-of-government protocols (rules & procedures)

Although the short-term response prioritized simplification to achieve rapid removal of the designation, a more integrated and strategic approach is necessary for long-term preparedness.

- Relevant ministries jointly formulated response measures during the handling of this case. However, as complex issues involving diplomacy, industry, and science and technology are intertwined, it is necessary to continuously strengthen the response system to leverage each ministry's strengths effectively through pre-established protocols (rules and procedures).
  - (Establishment of response protocols aligned with ministry missions) Consideration should be given to the missions and strengths of the Ministry of Foreign Affairs, Ministry of Trade, Industry and Resources, Ministry of Science and ICT, the National Intelligence Service, and other relevant ministries to allocate roles optimally and institutionalize protocols for rapid response in emergency situations.

    * This system should go beyond simple inter-ministerial coordination and function as a strategic response system for complex crises combining technology, diplomacy, and industry, designed around institutionalization, data-driven processes, and expert linkages.
  - (Expert committees and public-private participation structure) There is a need to establish sectoral committees for sensitive technologies handling high-risk areas and prepare a system to incorporate real-time feedback from industry and research institutions.

    ※ In the U.S., organizations such as the BIS, NSC, and CFIUS operate cooperative strategic frameworks. Similarly, Japan's NSC economic security office integrates diplomatic, industrial, and intelligence functions to respond comprehensively to sensitive issues. Additionally, METI is seeking to strengthen the intelligence system for industrial technology information by establishing public-private think tanks under its supervision (NHK, October 1, 2024; Ministry of Economy, Trade and Industry, October 2024).

Recommendation 3. (Preventive system) Establishing proactive monitoring mechanisms for key sanction measures such as sensitive countries and export controls

- The current domestic response system primarily operates at the stage of tracking "results," such as legal revisions or export controls announcements, limiting its ability to detect potential designations in advance and design policy measures proactively.

- Although various sanctions and restrictive measures have emerged sporadically, including the U.S. DOE's designation of South Korea as a sensitive country and the United States' announcement of reciprocal tariff measures in April, there are limitations to anticipating and responding to such developments in advance.
- Additionally, strategic export controls aimed at strengthening global competitiveness in the AI sector have been implemented, such as the Biden Administration's export control measures on high-performance computing equipment in November 2024 and January 2025, as well as the Trump Administration's export controls on H2O chips targeting China.[53]

The sensitive country designation should not be understood as a "sudden event" but as part of the ongoing international situation and technology control trends, which raises the need to establish a proactive, continuous monitoring system for early detection and risk anticipation.

- This monitoring system should be designed to detect issues promptly, analyze them comprehensively, and take policy responses based on collected information.
- (Continuous AI-based risk map) There is a need to quantify risk indices through sensitivity analysis of technology clusters by country and visualize them on maps to incorporate them in policy design.
- (Development of Early Warning Index) Potential designation can be predicted in advance by quantifying changes in the U.S. and EU control frameworks, legislative proposals, and public hearings as indicators.
- (Integration for private sector use) An open API can be provided to enable major industries and research institutions to develop their response plans based on the risk map.

  ※ The U.S. Center for Security and Emerging Technology (CSET) develops country-technology risk maps using AI to analyze publications, investments, and sanctions data.

53) Hankyoreh (April 16, 2025), "U.S. imposes indefinite restriction on NVIDIA's 'H2O chip' exports to China," https://www.hani.co.kr/arti/international/america/1192622.html (accessed: April 25, 2025)

※ The EU utilizes Tools for Innovation Monitoring (TIM) (TIM Dual-Use) to track research trends in sensitive technologies and apply it as a tool to assess control possibilities.

Recommendation 4. (Intelligence acquisition) Diversifying diplomatic channels and strengthening strategic intelligence

Insufficient diplomatic channels and science and technology expertise may lead to a decline in bargaining power in response to sensitive country designation.

- Science and technology diplomacy should no longer be viewed merely as a support tool for traditional diplomacy but as a central instrument for securing technological sovereignty and leading global technology standards.
- In particular, in cases such as the present issue-which require scientific-technical expertise and diplomatic resolution-securing experts* for response may become critically important.

  * For issues similar to this sensitive country designation, actual contextual information and intelligence, which cannot be obtained through official diplomatic channels, should be collected through networks of science and technology specialists and diplomatic/security experts.

- Additionally, when the interests of the relevant countries conflict on specific issues, diversification of diplomatic channels is necessary to address potential stalemates strategically in government-centered negotiations.

To address these issues, it is necessary to establish a diplomatic buffer strategy using unofficial channels, such as cooperation systems among think tanks in science, technology, and economic security in like-minded countries, as well as multilateral private-sector forums.

- (Intelligence Network Establishment) South Korea, Japan, selected EU countries, Australia, and Canada should establish a science, technology, and economic security intelligence network.
  - Cooperation among think tanks specializing in science, technology, and economic security in major like-minded countries should be used to collect and share diverse information beyond government peripheries, serving as a key axis

for formulating intergovernmental cooperation strategies.

- (Emergency Issue Response System Operation) In the event of designation, export restrictions, or investment blockages, urgent issue meetings should be convened to discuss joint response measures and provide recommendations to the government.
  - Collaboration with global think tanks can help develop evidence-based response strategies for pending issues.
- (Establishment of communication channels through private sector participation) These channels should serve as a private sector-government information bridge involving industrial actors (particularly companies holding sensitive technologies) and research institution experts.
- (Acquisition and use of non-public information channels) Based on intergovernmental and public-private trust, information acquisition and analytical support must be systemized through official channels and non-public pathways.
  - These channels should be used to share expanded contexts and information (intelligence) from various perspectives that cannot be collected by the government.

Recommendation 5. (Legal framework improvement) Strengthening the foundation for interagency collaboration on information diagnosis and analysis through emerging security legal framework improvement

The current sensitive country issue has highlighted the importance of intelligence-driven response to emerging security.

- The government has previously pursued the enactment and amendment of emerging security legislation, including economic security and science and technology security, in response to rapidly changing international affairs, such as the South Korea-Japan trade conflict, COVID-19, and conventional war outbreaks.

- Among these, continuous monitoring and management of domestic and international industrial, economic, supply chain, and science and technology information related to emerging security are more important than ever.

There is a need to organize the information diagnosis and analysis provisions within the emerging security legislation that each ministry has enacted or amended.

- (Fragmented information analysis and management provisions) While each ministry has competitively pursued the enactment and amendment of emerging security legislation, provisions that could be commonly utilized have become dispersed and fragmented, with each ministry advancing its policies. Provisions for information and intelligence analysis and management are an example of this fragmentation.
- (Revision of Information-Related Provisions in Emerging Security Legislation) There is a need to refine the legal frameworks administered by relevant ministries in the area of emerging security, either by strengthening linkages among separately conducted information analysis and management activities or by enacting specialized legislation primarily aimed at developing intelligence on science and technology, industrial technology supply chains, and related information.
  - Currently, emerging security measures are implemented under the Ministry of Science and ICT's Special Act on the Fostering of National Strategic Technology; the Ministry of Trade, Industry and Resources' Act on Special Measures for Strengthening the Competitiveness of and Protecting National High-Tech Strategic Industries; and the Ministry of Economy and Finance's and the Ministry of Trade, Industry and Resources' three acts on supply chains.*
  - It is necessary to supplement the knowledge- and information-management provisions in each legislation by linking their contents through a mutual referencing scheme or common provisions.
  - As information analysis and monitoring systems serve as the starting point for evidence-based policy implementation, there is a need to strengthen the corresponding policy focus.

- In this case, another option could be to establish a separate special law that links the respective functions currently pursued by each ministry, promoting interagency collaboration and enabling a more integrated response.

  * (Ministry of Science and ICT) Special Act on the Fostering of National Strategic Technology, Article 14 (Management of knowledge and information related to national strategic technology)
  (Ministry of Economy and Finance) Framework Act on Supply Chain Stabilization Support for Economic Security, Article 15 (Operation of early warning systems)
  (Ministry of Trade, Industry and Resources) Act on Special Measures to Strengthen Competitiveness and Stabilize Supply Chain of Materials, Components, and Equipment, Article 23-2 (Operation of early warning system for materials, components, and equipment supply chain)
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  (Ministry of Trade, Industry and Resources) Special Act on National Resources Security Chapter II, Article 8 (Comprehensive management of resources security information), Chapter 3 Preparation for Resources Security Crisis, Section 1 Establishment of Early Warning System for Resources Security Crisis

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